

Causal Impact of Old Age Security (OAS) Eligibility on Labor Force Participation for Seniors in Canada

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I. Introduction

With the steady and fast-aging population of Canada over the past two decades, the aging rate is expected to continue growing at a potentially faster pace and has drawn the importance of pensions in front of us; specifically, the rate has grown from 8 percent in 1971, 12 percent in 1996, 15% in 2016 and now has gone up to 18.9% in 2023 (Bélanger et al., 2016; Statistic Canada, 2023). With the continuously shifting demographic, understanding how retirement benefits affect seniors' labor market behavior is critical for both economic forecasting and policy development.

This paper specifically focused on the Old Age Security (OAS) – a functional, age and residency-based component of Canada's public pension system. We aim to assess the **causal effect of OAS eligibility on labor force participation for seniors in Canada**, with an analysis focused on immigrants. Considering OAS' clearly listed rule and cut-off: being an eligible Canadian citizen or legal resident aged above 65, with a minimum of 10 years residency in Canada after age 18 (Government of Canada, 2024).

Given our identities as international students and our interest in immigration policy and aging, we are particularly interested in immigrants and how their labor market behavior may vary based on OAS eligibility. We aim to provide new evidence on how retirement policies interact with demographic characteristics, influencing retirement decisions.

II. Background and Literature Review

Ambrozas et al. (2020) discussed the “three pillars” holding Canada’s retirement income systems:

- The first pillar provides “benefits based on age and years of residence in Canada”, including OAS, Guaranteed Income Supplement (GIS), Allowance, and Age Credits;
- The second one with “mandatory earnings-related programs”, including Canada Pension Plan/ Quebec Pension Plan (CPP/QPP);
- The third is “voluntary retirement income tools”, composed of Workplace Registered Pension Plans (RPPs) and private savings (Registered Retirement Savings Plans [RRSPs] and Tax-Free Savings Accounts [TFSAs]).

While researches have been done on broad impacts and influences of these systems, limitation remains in empirical evidence on causal effect of eligibility of a single program (OAS specifically) on retirement decisions. Past studies have explored related themes including the general effects of public pension structure alter the incentives to retire, focusing on how health, private pensions, and spousal dynamics shape the decision (Schirle, 2006); determinants (e.g., financial, health and social factors) of retirement age and labor force participation for senior workers using descriptive approach (Belanger et al., 2016); and labor market attachment among seniors in Canada and the United States focusing on part-time, involuntary job separation, and cognitive outcomes (Campbell, 2023).

Shown the potential correlation with the pillars with factors such as the health status emphasized the importance of investing related policy impact, together given Canada's increasing reliance on immigration as a policy response to an aging population and labour shortages (Esses et al., 2022), our study aims OAS eligibility as a quasi-experimental

treatment, using citizenship and immigration time as the key identification variables, proposing that immigrants who arrive in Canada that have not met the 10-year residency requirement forms a natural control group opposing Canadian-born or long-term resident immigrants on eligibility. This allows us to fill the notable gap in existing literature on the effect of OAS on seniors' labor force participation behaviors.

III. Data

The source of our dataset is the 2021 Canadian Census (Statistics Canada, 2023). During the data cleaning process, echoing our research question, we took a sample that only includes people aged from 50 to 74 (numbered 13-19 in "*AGEGRP*", grouped age), as our analysis focuses on those who are seniors. We excluded the "never worked" observations and missing values in labor force status (*LFACT*). Additionally, we retained only individuals whose immigrant status is classified as 1-citizen, or 2-immigrant in *IMMSTAT* (immigration status) to preserve the cleanliness of our data.

Based on the information on Labour force status, we created our outcome variable, ***Retired (Yi)***, as a binary indicator: it takes the value 1 for those described with "Not in the labor force" and 0 otherwise. To capture OAS eligibility, we created two additional dummy variables: one indicating whether the individual has lived in Canada for at least 10 years (*eligibility*), taking 1 for *YRIM* $\geq$ 2011 and 0 otherwise, and another indicating whether the individual is 65 years old or older (*post*), taking 1 for "*AGEGRP*" $\geq$ 17 and 0 otherwise.

We then further divided the sample into citizen-immigrant and immigrant-only subsets, to better control the confounding covariates in our causal analysis. Finally we focus on the following variables: ***AGEGRP***, a categorical variable of age group; ***LFACT***, Labour: a categorical variable of Labour force status - Detailed; ***YRIM***, Immigration: a categorical

variable of Year of immigration; *IMMSTAT*, Immigration: a binary variable of Immigrant status; *Retired*, a dummy variable indicating whether or not retired in 2021; *eligibility*, a dummy variable indicating whether or not having been in Canada for 10 years or longer; *post*, a dummy variable indicating whether or not being aged 65 or older in 2021.

Below (Table 1.1 & 1.2) are the summary statistics for our main variables of interest in the citizen-immigrant dataset and immigrant-only dataset, respectively:

AGEGRP	LFACT	YRIM	IMMSTAT	Retired	eligible	post
Min. :13.00	Min. : 1.000	Min. : 1	Min. :1.00	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:14.00	1st Qu.: 1.000	1st Qu.:2011	1st Qu.:1.00	1st Qu.:0.0000	1st Qu.:1.0000	1st Qu.:0.0000
Median :16.00	Median : 1.000	Median :9999	Median :1.00	Median :0.0000	Median :1.0000	Median :0.0000
Mean :15.62	Mean : 6.125	Mean :7453	Mean :1.28	Mean :0.4088	Mean :0.9731	Mean :0.3296
3rd Qu.:17.00	3rd Qu.:13.000	3rd Qu.:9999	3rd Qu.:2.00	3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:1.0000
Max. :19.00	Max. :13.000	Max. :9999	Max. :2.00	Max. :1.0000	Max. :1.0000	Max. :1.0000

Table 1.1: the summary statistics of the citizen-immigrant dataset

AGEGRP	LFACT	YRIM	IMMSTAT	Retired	eligible	post
Min. :13.00	Min. : 1.000	Min. : 1.0	Min. :2	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:14.00	1st Qu.: 1.000	1st Qu.: 6.0	1st Qu.:2	1st Qu.:0.0000	1st Qu.:1.0000	1st Qu.:0.0000
Median :15.00	Median : 1.000	Median : 9.0	Median :2	Median :0.0000	Median :1.0000	Median :0.0000
Mean :15.51	Mean : 5.729	Mean : 894.8	Mean :2	Mean :0.3691	Mean :0.9037	Mean :0.3152
3rd Qu.:17.00	3rd Qu.:13.000	3rd Qu.:2003.0	3rd Qu.:2	3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:1.0000
Max. :19.00	Max. :13.000	Max. :8888.0	Max. :2	Max. :1.0000	Max. :1.0000	Max. :1.0000

Table 1.2: the summary statistics of the immigrant-only dataset

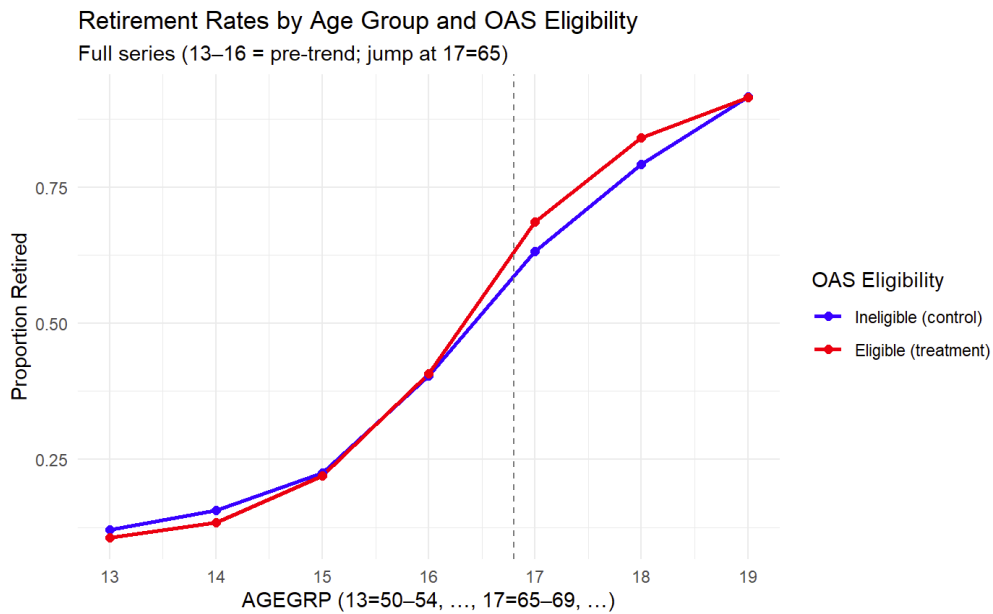
IV. Methodology

To estimate the causal impact of OAS eligibility on retirement behavior in Canada, we employed a Difference-in-Differences (DD) framework. The outcome variable is a binary indicator of retirement status, *Retired* (Y_i), and the treatment is defined as being **above the OAS eligibility age (65) and living in Canada for at least 10 years (Di)**. The time dimension is the age groups the observations belong to, and in the group dimension, we compare the difference in eligible and ineligible groups.

Note here, although born Canadians and legal residents who have stayed in Canada for 10 years who have not reached age 65 are, strictly speaking, not eligible for the OAS yet; they are “potentially eligible” once they are qualified by their age (an “automatic” process), they are eligible for the program; with data for each individual’s status across years missing due to limitation of the dataset, the common trend assumption (CTA) needed for the DD analysis can only be measured when considering the eligible=1 group not only contains those who are already eligible (aged ≥ 65) but also those who are “potentially eligible”, as we described before.

Our analysis is conducted in two parts. First, we compared all Canadian citizens and legal residents to recent immigrants using a DD approach that leverages the fact that recent immigrants are not qualified for OAS even after turning 65, serving as a natural control group. Second, aiming to reduce potential heterogeneity between groups, we restricted the sample to immigrants only and again compared those eligible versus ineligible for OAS before and after age 65. With the sub-grouping process existing, we did not add other covariates to our DD model; instead, we focused on the immigrant subgroup in which the treatment and control groups share more common characteristics, considering the intuitive potential difference embedded between born-citizens and immigrants.

To interpret our estimates as causal, the common trends assumption is required, that in the absence of OAS eligibility, retirement behavior would have evolved similarly across the treated and control groups across age before the eligible group becomes treated. We partially validated this by comparing pre-treatment trends across groups in Figure 1.1 and Figure 1.2. Not only using the “eyeball test”, analysis for the pre-trend groups is conducted. For the general group containing both born-citizens and immigrants, all the interaction terms are insignificant while the coefficients are small (close to 0); a bit of variation exists but given the limitation of our dataset, we assume the common CTA holds for this group.



```
lm(formula = Retired ~ eligible * factor(AGEGRP), data = pre_df)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.4078	-0.2195	-0.1340	-0.1061	0.8939

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.120488	0.006418	18.774	< 2e-16 ***
eligible	-0.014374	0.006640	-2.165	0.030393 *
factor(AGEGRP)14	0.035437	0.010351	3.424	0.000618 ***
factor(AGEGRP)15	0.104512	0.012563	8.319	< 2e-16 ***
factor(AGEGRP)16	0.283099	0.014799	19.130	< 2e-16 ***
eligible:factor(AGEGRP)14	-0.007552	0.010619	-0.711	0.476982
eligible:factor(AGEGRP)15	0.008905	0.012771	0.697	0.485605
eligible:factor(AGEGRP)16	0.018636	0.014980	1.244	0.213473

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3983 on 251995 degrees of freedom

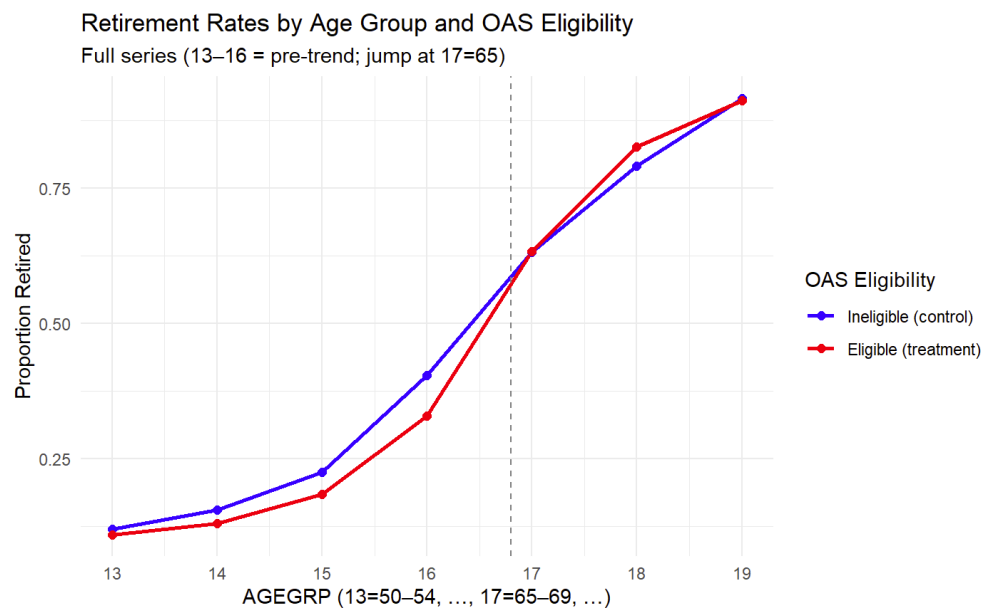
Multiple R-squared: 0.07992, Adjusted R-squared: 0.07989

F-statistic: 3127 on 7 and 251995 DF, p-value: < 2.2e-16

Figure 1.1: trends in retirement rate for citizen-immigrant group

When looking at the immigrants-only group, we see more variation for the pre-trend; echoing with the graph, the slopes for the two interaction terms get more and more significant; this means the CTA here holds not as strong as the general group; however, we observe the slope of the coefficients turns negative during the pre-trend, but dramatically

shifts back to positive according to the graph at *AGEGRP* 17, which also shows a sudden increase in retirement rate for the eligible group.



```
lm(formula = Retired ~ eligible * factor(AGEGRP), data = pre_df_imm)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.4036	-0.1842	-0.1304	-0.1097	0.8903

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.120488	0.006110	19.719	< 2e-16	***
eligible	-0.010776	0.006857	-1.572	0.116043	
factor(AGEGRP)14	0.035437	0.009855	3.596	0.000323	***
factor(AGEGRP)15	0.104512	0.011960	8.738	< 2e-16	***
factor(AGEGRP)16	0.283099	0.014090	20.093	< 2e-16	***
eligible:factor(AGEGRP)14	-0.014763	0.010740	-1.375	0.169275	
eligible:factor(AGEGRP)15	-0.029998	0.012692	-2.364	0.018104	*
eligible:factor(AGEGRP)16	-0.063217	0.014766	-4.281	1.86e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3792 on 71991 degrees of freedom
Multiple R-squared: 0.04633, Adjusted R-squared: 0.04624
F-statistic: 499.6 on 7 and 71991 DF, p-value: < 2.2e-16

Figure 1.2: trends in retirement rate for immigrant-only group

Furthermore, we assume there are no simultaneous policy changes or differential shocks affecting one group but not the other at the cutoff age. Lastly, we assume correct specification in our DD models exclude the influence of unobserved confounders correlated

with both eligibility and retirement. By conducting different linear model specifications and analyzing different comparison groups, we aim to strengthen the credibility of our causal interpretation.

Model Specification

Our models include both baseline (additive) specifications and DD models with interaction terms to capture treatment effects. We estimated each specification using both Linear Probability Models (LPMs) and logistic regressions for robustness.

- Baseline Specification

We began with an additive model for both the citizen–immigrant dataset and the immigrant-only sample:

$$Retired_i = \beta_0 + \beta_1 Post_i + \beta_2 Eligible_i + \epsilon_i$$

This model estimates the independent associations of being over age 65 (*Post*) and being eligible for OAS (*Eligible*) on retirement status, without accounting for interaction effects.

- Difference-in-Differences Specification

To identify the causal effect of OAS eligibility, we included an interaction term between *Post* and *eligible*:

$$Retired_i = \beta_0 + \beta_1 Post_i + \beta_2 Eligible_i + \beta_3 (Post_i \times Eligible_i) + \epsilon_i$$

Here, the coefficient β_3 captures the DD estimate — the change in retirement probability after age 65 specifically for those eligible for OAS, relative to those who are not.

This model is appropriate for testing whether OAS eligibility causes an increase in retirement rates.

We estimated this specification using both: 1. A Linear Probability Model (LPM) for direct interpretability of coefficients; 2. A Logistic Regression accounts for the binary nature of the dependent variable and avoids the drawbacks of LPMs, such as predicted probabilities outside the $[0,1]$ range.

We ran all specifications on both the full citizen–immigrant sample and a restricted immigrant-only sample to test the robustness of our findings and reduce the influence of group-level heterogeneity. In all cases, the specification appropriately addressed our research question by isolating the interaction between policy eligibility and age threshold, which defined treatment in our context. The models are clearly aligned with our DD identification and enable causal interpretation under our assumptions.

V. Results

1. Citizen–Immigrant Sample

As shown in Table 2.1, columns (1) and (2) suggest that both Post ($\text{age} \geq 65$) and eligible are individually associated with significantly higher retirement rates. However, the coefficient on the interaction term ($\text{Post} \times \text{Eligible}$) in both the linear and logistic models is small and statistically insignificant (0.003 in LPM and -0.012 in logit). This implies that once we control for general age and eligibility effects separately, becoming OAS-eligible does not significantly increase retirement rates among those already likely to retire due to age alone. The insignificant interaction term interprets that in the common group, being eligible to OAS

increase the eligible odds' likeliness of retirement by 0.3% for the linear model comparing to the control group, while decrease the likeness of retirement by 1.22% for the for the odds of retirement at the eligibility cutoff in the logistic model.

Citizen-Immigrant Models: OAS Elibility Effects			
	<i>Dependent variable:</i>		
	Retired		
	<i>OLS</i>		<i>logistic</i>
	Baseline	DID (LPM)	DID (Logit)
	(1)	(2)	(3)
Post	0.568***	0.564***	2.590***
Eligible	0.047***	0.046***	0.289***
Post × Eligible		0.003	-0.012
Constant	0.176***	0.177***	-1.538***
Observations	375,910	375,910	375,910
R ²	0.296	0.296	
Adjusted R ²	0.296	0.296	
Log Likelihood			-196,840.500
Akaike Inf. Crit.			393,688.900
<i>Note:</i>			
Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1			

Table 2.1: results of citizen-immigrant models

2. Immigrant-Only Sample

In comparison, the results are notably different in the immigrant-only sample (Table 2.2). The linear probability model in column (2) shows a statistically significant positive effect of OAS eligibility on retirement ($\text{Post} \times \text{Eligible} = 0.020, p < 0.1$), and the logistic regression in column (3) also shows a positive (though not statistically significant put still significant under the 0.1 level) coefficient (0.098). This suggests that among immigrants, OAS eligibility may play a more direct and important role in the decision to exit the labor force. Specifically in the immigration only group, a bit more significantly, being eligible to OAS increase the eligible odds' likeliness of retirement by 2% for the linear model

comparing to the control group, while increase the likeness of retirement by 10.3% for the for the odds of retirement at the eligibility cutoff in the logistic model.

These results align with our theoretical expectations: OAS, as a foundational benefit not tied to prior labor market participation, may serve as a stronger retirement incentive for recently arrived immigrants.

Immigrant-Only Models: OAS Eligibility Effects			
	<i>Dependent variable:</i>		
	<i>Retired</i>		
	<i>OLS</i>		<i>logistic</i>
	Baseline	DID (LPM)	DID (Logit)
	(1)	(2)	(3)
Post	0.583 ^{***}	0.564 ^{***}	2.590 ^{***}
Eligible	0.013 ^{***}	0.009 ^{**}	0.062 ^{**}
Post × Eligible		0.020 [*]	0.098
Constant	0.174 ^{***}	0.177 ^{***}	-1.538 ^{***}
Observations	105,133	105,133	105,133
R ²	0.316	0.316	
Adjusted R ²	0.316	0.316	
Log Likelihood			-52,369.430
Akaike Inf. Crit.			104,746.900
<i>Note:</i> [*] p<0.1; ^{**} p<0.05; ^{***} p<0.01 Standard errors in parentheses. ^{***} p<0.01, ^{**} p<0.05, [*] p<0.1			

Table 2.2: results of immigrant-only models

VI. Discussion & Extension

Our findings provide multiple evidence on the causal effect of OAS eligibility on retirement behavior. While we do not find a significant effect in the general sample, possibly due to confounding retirement incentives like CPP, the immigrant-only linear probability models reveal a modest but significant increase in retirement probability following OAS

eligibility, by 0.02. This effect size is too modest to conclude the economic significance of the findings even when statistically significant.

Moreover, with the coefficient of 0.098 on the interaction term in the logistic regression for the immigrant-only sample, we see OAS eligibility increases the odds' likelihood of retirement by 10.3% (with $e^{0.098} \approx 1.103$) for eligible immigrants aged above 65, relative to ineligible immigrants. However, the effect is not statistically significant, so some assumptions could be helpful for it to be meaningful, such as the base retirement rate in the group is nontrivial. For example, it means a 5% increase in the base retirement rate of 50%.

Other evidence of causal robustness includes the validity of our Common Trends Assumption (CTA) and the consistency of signs and magnitudes across specifications. In the methodology section, the CTA appears more credible for the citizen-immigrant sample, but relatively less for the immigrant-only sample, potentially due to greater heterogeneity in retirement behavior among immigrants. Additionally, while the estimated causal effects differ in sign across the two samples (-0.012 vs. 0.098), neither is statistically significant (though the immigrant-only group has a p-value of 0.1352, much better than the general group where it is 0.85). This divergence may reflect underlying differences in how OAS eligibility influences labor supply decisions across groups, but it also highlights the need for cautious interpretation of point estimates in the absence of strong statistical significance.

- Limitations & Extensions:

As partially mentioned before, limitations exist in the study that obstruct us from reaching more significant/ generalizable results. First, the dataset we are using here uses cross-sectional data from a single year to a large extent limiting the possibility of observing changes through years (the real-time effect in DD) while the grouped age also holds us back

from more continuous/ focused analysis. Future studies in this area could try finding more detailed data on the specific topic across years, creating a more robust dataset that could lead to more accurate and reliable results. Second, our aim of focusing our analysis on OAS could easily be influenced by other confounding variables, with the best example being CPP, that also the eligibility rule is largely different and more complex, but it still provides retirement income similar to OAS, being a strong covariate; however, even eliminating the impact of such covariates, we may not see an obvious increase in the explanatory power of the model, considering the complex demographic in Canada (which is the reason why we find our R-squared around 0.4 to be solid). While controlling such covariates is always challenging and could be directions that future research can focus on, the insignificance issue of the interaction terms cannot be solved for now which may cause questions about the credibility of our model; this is why other tests may help solidify our analysis as a causal relationship.

To further solidify the causal analysis, there is a potential extension by using the Regression Discontinuity (RD) design. If we had access to panel data with continuous age rather than age group bins, we could implement a sharp RD design centered at the OAS eligibility threshold (age 65, focusing on e.g., 64-66). The method allows us to more credibly isolate the causal effect on retirement behavior by comparing the retirement rates just below and just above the threshold, where other covariates are assumed as almost the same in this subset around this cutoff. RD would relax the requirement of CTA, which is not strongly credible with our dataset (shown in Figure 1.1 and 1.2 in the Methodology section), in the DD approach. Therefore, this approach would help us address the potential failure of one essential assumption with DD models, and it leverages the deterministic eligibility rule of OAS in age as a natural cutoff for causal identification and the typically sharp changes in labor force participation around retirement age.

VII. Conclusion

In this paper, we use the eligibility rule of Canada's Old Age Security (OAS) program, a cutoff at age 65 for citizens and immigrants who have lived in Canada for at least 10 years, to identify the causal impact of retirement benefits on seniors' labor force participation. We compared the retirement rate regarding the eligibility of OAS using a Differences-in-Differences framework, among both the full sample of Canadian-born and long-term immigrants and the immigrants-only group. While the age 65 jump in retirement is relatively visible in the trend graphs, the causal effect of OAS eligibility itself is small and statistically insignificant in the full sample. In contrast, for the immigrants-only group, the OAS eligibility interaction is positive and weakly significant, suggesting that immigrants who stayed in Canada for a relatively shorter period are more likely to be impacted by the retirement income provided on their labor force participation behavior.

The paper makes two key contributions. First, we provide a quasi-experimental estimate of OAS eligibility on retirement decisions using cross-sectional census data, isolating the effect of a single pillar of Canada's three-pillar system. Second, we highlight the heterogeneous effects of social security programs across different demographic groups, particularly by immigration status and age of arrival. From a policy perspective, the results suggest that OAS may play a more critical role for individuals with limited access to other retirement income sources, such as recent immigrants. These insights indicate the importance of more targeted policy designs focusing on different demographic groups and supporting ongoing discussions around the amount of total pension income, equity, and budget allocation.

A promising direction for future research would be to collect or access data with continuous age and panel structure with real time factors across different years. This would

allow for a regression discontinuity design to more precisely isolate the timing and dynamics of labor force exits around retirement age. More detailed data would also enable a closer analysis of anticipation effects, earnings histories, and spillovers from other income sources like CPP or GIS. Together, these extensions would further strengthen the causal interpretation and policy relevance of the findings.

We hope this paper builds a solid foundation on OAS' sole impact on labor force participation behavior among Canadian seniors, especially among immigrants, and hope to conduct future research with more comprehensive, robust analysis to generate more credible and reliable results, making our own contributions.

AI / LLM Use Disclosure

In accordance with Sage's policy on the use of generative AI tools (SAGE Publications Ltd., 2023) and the course AI guidelines, we hereby disclose that we used the following AI-assisted tools strictly as follows:

- **ChatGPT (OpenAI, March 14 Version)** was employed to generate search text for generating research ideas, adjusting codes during plotting and final formatting of tables, and assessing the alignment of the paper to the rubric of the course. No substantive text (e.g. introduction, literature summaries, methodological descriptions, or results interpretation) was generated by ChatGPT; all ideas, analyses, and write-ups are the authors' own.
- **Grammarly** was used to refine grammar.

We did not use AI tools to produce original content, draft analysis, or interpret results. All factual checks, literature research and review, modeling decisions and interpretations were conducted by the authors.

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